

triple_diff

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```
aggregate_data <- function(data,
                           cols_to_keep,
                           pop_col,
                           agg_noneuca = TRUE){

  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(state, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("state", "region"))

  # aggregate regions if requested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(pick(-c(n_arrests, pop))) %>%
      summarize(n_arrests = sum(n_arrests),
                pop = sum(pop))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(state, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # population map
  pop_map <- data_pop %>%
    group_by(state, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")

  # checksum value
  checksum <- data_pop %>%
```

```

    group_by(state, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")

# aggregate arrests, join population back in
data_pop <- data_pop %>%
  group_by_at(cols_to_keep) %>%
  summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
  left_join(pop_map, by = geo_keys) %>%
  mutate(ratio = n_arrests/pop*1000,
         log_pop = log(pop))

# checksum
total_aggregated <- data_pop %>%
  ungroup() %>%
  group_by_at(geo_keys) %>%
  summarize(pop = first(pop), .groups = "drop") %>%
  pull(pop) %>%
  sum(na.rm = TRUE)

total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)

if (!(dplyr::near(total_checksum, total_aggregated))) {
  warning("Checksum Failed!")
}

return(data_pop)
}

```

```

## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
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## [1] 6.247311 2.529624 9.964997

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```

```
## [1] 0.07399809
```

```
## [1] 1.6511291 -0.1602262 3.4624843
```